

Akshay D Shetty

akshay18shetty@gmail.com

www.linkedin.com/in/akshay18shetty

www.insightbyakshay.in

EDUCATION

Master's in Data Science (AI/ML)

Manipal Institute of Technology, Manipal

August 2023 – Present

Bachelor's in Computer Science and Engineering

SMVITM

August 2022

TECHNICAL SKILLS

Machine Learning & AI:

PyTorch, TensorFlow, Keras, Scikit-learn, Computer Vision, NLP

Programming Languages:

Python (Advanced), SQL, C, HTML/CSS

Data Science:

Pandas, NumPy, Matplotlib, Data Analysis, Statistical Modeling

Tools & Databases:

MySQL, Git, Jupyter Notebooks

WORK EXPERIENCE

Data Science Intern | Labmentix

March 2025 – May 2025

- Worked on multiple data science projects involving exploratory data analysis (EDA), data cleaning, and feature engineering.
- Developed interactive data visualizations to uncover insights and support model development.
- Built and evaluated machine learning models using Python and industry-standard libraries.
- Collaborated with the team to present findings and improve model performance iteratively.

PROJECTS

Multi-Crop Plant Leaf Nutrient Deficiency Classification

- Developed an ensemble deep learning model using EfficientNet, ResNet, and DenseNet to classify nutrient deficiencies in banana, citrus, and rice crop leaves with over **90% accuracy**.
- Applied **transfer learning**, **data augmentation**, and **fine-tuning** to improve performance on limited, imbalanced datasets.
- Integrated **Grad-CAM** for visual explanation of predictions, enhancing model interpretability for agricultural experts.
- Utilized custom PyTorch pipelines and ensemble averaging to boost cross-crop generalization.
- Technologies: PyTorch, CNN, Transfer Learning, Ensemble Learning, Grad-CAM, OpenCV, NumPy, Matplotlib

Speech Emotion Recognition

- Developed an end-to-end emotion recognition system achieving 85% accuracy on diverse audio datasets.
- Performed audio pre-processing, noise reduction, and feature engineering for emotion classification.
- Conducted comparative analysis of multiple ML algorithms, optimizing for both accuracy and processing speed.
- Technologies: Python, Scikit-learn, MFCC, Random Forest, SVM, Librosa.

CERTIFICATES

IBM Data Science: Coursera

Skills: Data analysis, Machine Learning, Data Visualization

December 2023

IBM Data Engineering: Coursera

Skills: Data pipelines, ETL processes, Big Data

April 2024